AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT249
Practical modern statistics

Final Examination
Fall Semester 2025/2026

Date:14 /Jan./2026

Number of Exam Pages: 6 (including this cover sheet)	**Time Allowed:** 120 Minutes.

Instructions:

- Total Marks: **100**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part1: Q-1 [20 marks]

A- [12 marks] Answer the following 5 questions as True or False (2 marks) and write the correct statement if the answer is false or state why it is false(1mark):

1- A time series is a sequence of observations measured over time.

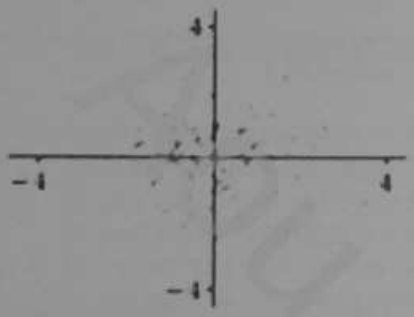
2- For a variable X, the between-groups variance V_b summarizes the spread of the observations on X within each group.

3- The forecast is a statement about the future value of a variable of interest.

4- The correlation matrix is used to describe the strength of relationships between variables in a multivariate data set.

5- In the data set in Figure**1** there is a clear positive relationship.

Figure 1



B- [8 marks]Fill in the blanks in the following 8 statements:

1- The ..... is the variable approximating a data set of any dimension by a single variable.
2- The cumulative percentage ..... is a measure of the quality of approximations consisting of two or more components.
3- Two approximations Y_1 and Y_2 to a multivariate data set are equivalent if constants $c_1 \neq 0$ and c_2 can be found such that .....
4- A measure of separation between two groups takes the value zero only if the ..... for the two groups are equal.
5- The differences (in absolute value) between the group means and the grand mean are combined into a single quantity, called the .....
6- A time series X_t is ..... in mean if it has constant mean, $E(X_t) = \mu$
7- For dimension reduction of p variables, if the variables are measured in different units, it is essential to ..... the variables.
8- The technique for finding linear combinations D that ..... the groups as much as possible is called canonical discrimination.

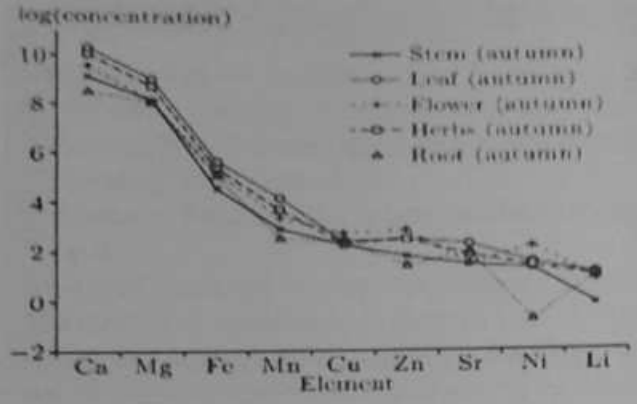
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Part2: Each question is worth 20 marks.
Answer the following questions:

Q–2: [4 +4 +4 +4 +4 marks]:

(a) A profile plot of the log concentrations of trace chemical elements found in five parts of the plants in the autumn -harvested plants is shown in Figure 2

Figure 2



1- Briefly describe the main features of this profile plot (Which 2 elements were found with higher concentrations? And Which element was found with lowest concentrations? Which 2 parts of the plant have the lowest concentration of most elements? Which one has the highest concentration of most elements.)
2- For which chemical elements does the pattern of the relative concentrations in the parts of the plant appear to differ from the pattern for all the other chemical elements? Explain your answer.

(b) Details of two hypothetical data sets are given below. Each data set comprises observations on 1000 individuals and has three variables. In each case, explain whether it would make sense to attempt to interpret a covariance matrix calculated from the unstandardized variables.

1- The three variables correspond to the weights of the individuals on three successive occasions.
2- The three variables correspond to the weekly income of an individual, the age of an individual, and the number of years the individual has had in full-time education.

(c) For the following matrix, state why it cannot be the correlation matrix for a set of data.

$$\begin{pmatrix} 1 & 0.2 & -0.3 & 0.7 \\ 0.2 & 1 & 0.4 & 0.5 \\ -0.3 & 0.4 & 1 & 0.9 \end{pmatrix}$$

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Q–3: [4 +4 +4 +4 +4 marks]:

(a) Five variables, $Y_1, Y_2, \ldots, Y_5$, were recorded for a set of 20 observations. The sample correlation matrix, calculated using these data, is

$$\begin{pmatrix} 1 & 0.77 & 0.09 & -0.74 & -0.70 \\ 0.77 & 1 & 0.13 & -0.80 & -0.76 \\ 0.09 & 0.13 & 1 & -0.17 & -0.23 \\ -0.74 & -0.80 & -0.17 & 1 & 0.78 \\ -0.70 & -0.76 & -0.23 & 0.78 & 1 \end{pmatrix}$$

Use this correlation matrix to answer the following questions.

1- Between which pair of variables does there appear to be the strongest linear relationship? Between which pair of variables does there appear to be the weakest linear relationship?

2- Describe the relationships between the variables implied by this correlation matrix ?

(b) A botanist measured various characteristics of plants from a particular species. The number of variables were reduced to be further analyzed by using principal component analysis (PCA), applied to the data after the variables had been standardized. Some information about the variables is given in Table 1.

Table 1

Variable	Units of measurement	Mean	Standard deviation	Loadings of first principal component	Value for Plant A	Value of standardized variable for Plant A
Weight of first seed	mg	469	5.2	0.8	470	0.192
Age	days	153	4.6	0.1	150	−0.652
Height	cm	30.5	3.1	−0.3	35	?
Number of leaves		93	3.9	0.5	90	−0.769

1- Give one reason why, in this case, it was important to standardize the data prior to using PCA.

2- Complete Table 1 by calculating the value of the standardized variable for height for Plant A.

3- Calculate the value of the first principal component for Plant A using the loadings given in Table 1.

Q–4: [4+4+4+4+4 marks]:

(a) Descriptions of two multivariate data sets are given below. For each data set, state the dimension p and the number of observations n.

1- In a study on interest rates in different countries, data on ten countries were collected. For each country, two interest rates were recorded: the average three-month interest rate during 2004, and the average interest rate for ten-year government bonds in 2004.

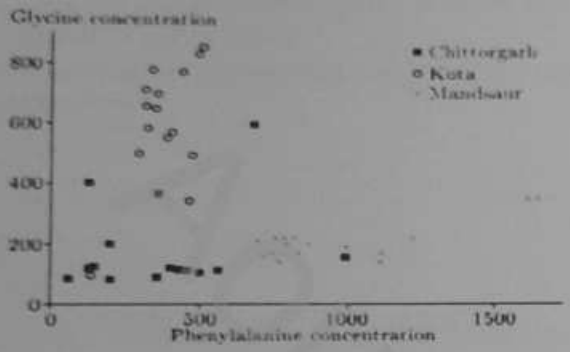
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2- A study was conducted to investigate the effect of exercise on the bones of people with osteoporosis. Samples of bone tissue from thirteen patients were studied. The levels of each of five proteins in each sample were measured.

(b) Figure 3 depicts a scatterplot of the concentrations (in g% of total amino acids) of two of the amino acids, glycine and phenylalanine samples of cultivated opium. Fifty of the samples came from three divisions: Chittorgarh (15 samples), Kota (17 samples) and Mandsaur (18 samples).

1- Which division of the cultivated opium samples have higher glycine concentrations? Which of them have higher concentrations of phenylalanine?
2- Which division of the cultivated opium samples can be separated by glycine concentration from others? And which division can be separated by phenylalanine concentration from others?
3- Which samples' locations tend to have similar glycine concentrations?

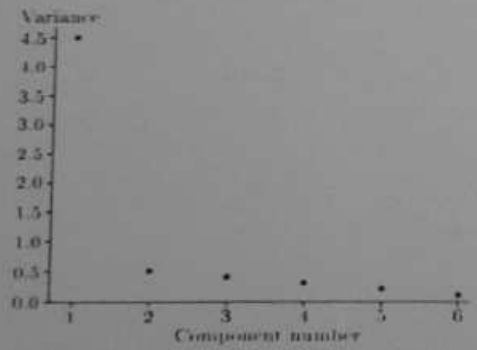
Figure 3



Q–5: [4+4+4+4+4 marks]:

(a) A principal component analysis was carried out on a standardized data set consisting of 96 observations and 6 variables. The resulting scree plot is shown in Figure 4.

Figure 4



1- Based on the scree plot given in Figure 4, how many components should be retained explain your answer.
2- Use Figure 4 to estimate the percentage variance explained by the first component.

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(b) There are five machines in a factory. Of these machines, (four) are working properly and (one) is defective. Machines which are working properly produce articles, each of which has independently a probability of (0.1) of being imperfect. For the defective machines this probability is (0.3).
(six) articles are examined

1- Find the probability that (2) articles out of the (6) articles are imperfect given that the machine is defective.
2- Find the probability that (2) articles out of the (6) articles are imperfect given that the machine is not defective.

A machine is chosen at random and (six) articles produced by the machine are examined.

3- What is the probability that the machine chosen is defective given that, of the (six) articles examined, two are imperfect and (four) are perfect?
Let D be "2 out of 6 imperfects", let M be "machine defective" and let $\bar{M}$ be "machine not defective."

Let D be "2 out of 5 imperfects", let M be "machine defective" and let $\bar{M}$ be "machine not defective."

End of questions

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